

Prime Time

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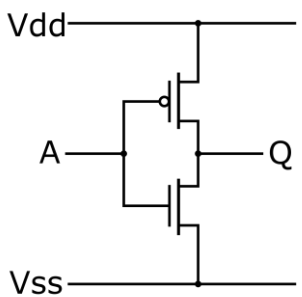
Outline

1. Introduction
2. Goal
3. Manual

1. Introduction

Hierarchy	Int Power	Switch Power	Leak Power	Total Power	%

- Total Power = Leakage Power + Dynamic Power
  - **Leakage(static) Power** : Power consumed by sub-threshold and gate leakage currents within transistors, even when they are in a static "off" state.
  - **Dynamic Power**
    - Switching Power : Power consumed by the charging and discharging of the external load capacitance only when the cell's output signal transitions.
    - **Internal Power**
      - Short-Circuit Power : Transient current flowing from VDD to GND when both PMOS and NMOS networks are momentarily "on".
      - **Internal Capacitance** : Power from charging/discharging parasitic capacitances on internal nodes within the cell.



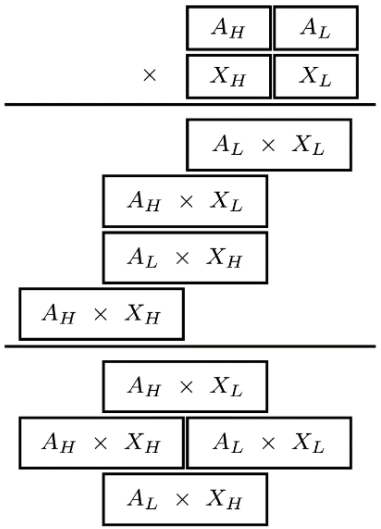
2. Goal --- average power report

- without clock gate

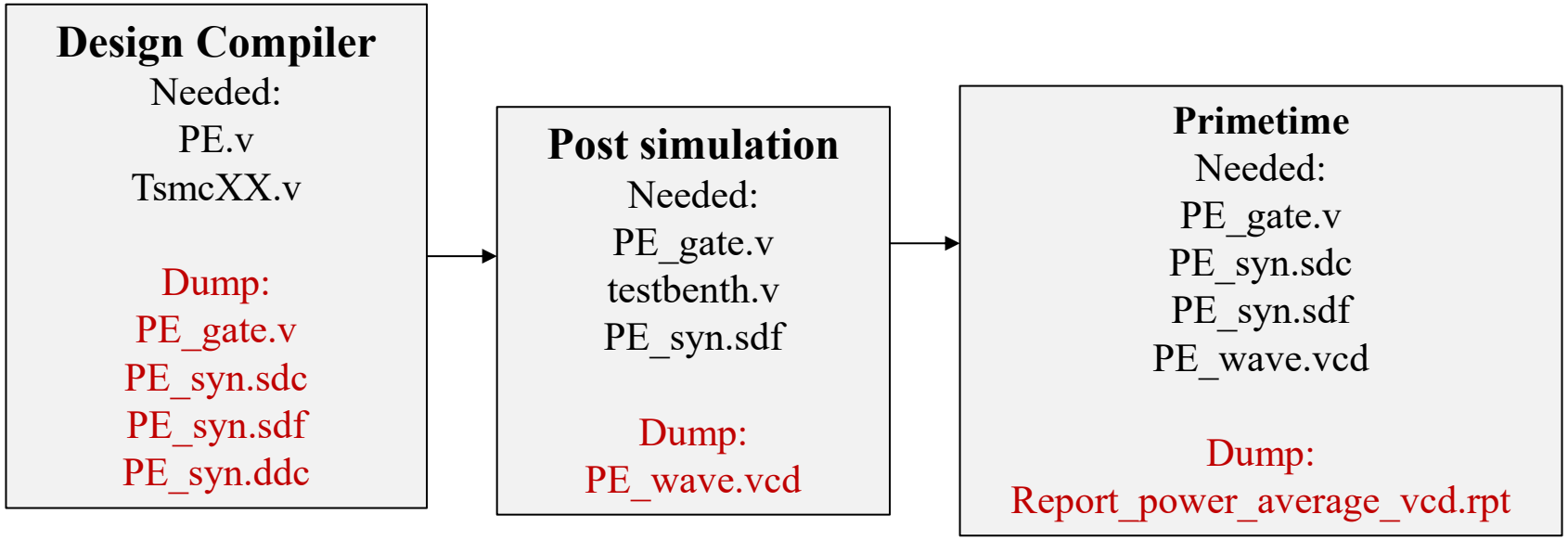
Hierarchy	Int Power	Switch Power	Leak Power	Total Power	%
top	6.08e-04	4.38e-04	2.28e-04	1.27e-03	100.0
a1_h (adder_2)	2.65e-05	6.40e-05	1.63e-05	1.07e-04	8.4
r328 (adder_2_DW01_add_0)	5.93e-06	5.53e-06	2.38e-06	1.38e-05	1.1
r329 (adder_2_DW01_sub_0)	6.53e-06	6.60e-06	2.89e-06	1.60e-05	1.3
r330 (adder_2_DW_cmp_0)	5.26e-07	2.83e-06	4.73e-07	3.82e-06	0.3
a2_h (adder_1)	4.82e-06	5.34e-06	2.81e-06	1.30e-05	1.0
r328 (adder_1_DW01_add_0)	4.36e-06	3.02e-06	2.14e-06	9.52e-06	0.7
r329 (adder_1_DW01_sub_0)	5.93e-06	5.53e-06	2.38e-06	1.38e-05	1.1
r330 (adder_1_DW_cmp_0)	4.36e-06	3.02e-06	2.14e-06	9.52e-06	0.7
a3_h (adder1_1)	9.66e-07	2.78e-08	5.16e-08	1.05e-06	0.1
r328 (adder1_1_DW01_add_0)	4.82e-06	5.34e-06	2.81e-06	1.30e-05	1.0
r329 (adder1_1_DW01_sub_0)	4.36e-06	3.02e-06	2.14e-06	9.52e-06	0.7
r330 (adder1_1_DW_cmp_0)	6.87e-06	6.51e-06	2.12e-06	1.55e-05	1.2
p0 (pipeline0_1)	1.32e-04	1.50e-05	6.91e-06	1.54e-04	12.1
p2_1_h (pipeline2_1)	3.11e-05	9.22e-07	2.14e-06	3.42e-05	2.7
p1 (pipeline1_0)	2.98e-05	3.30e-06	1.65e-06	3.48e-05	2.7
p2 (pipeline1_7)	3.02e-05	2.96e-06	1.61e-06	3.48e-05	2.7
p3 (pipeline1_6)	2.97e-05	3.01e-06	1.57e-06	3.42e-05	2.7
p4 (pipeline1_5)	3.00e-05	3.61e-06	1.68e-06	3.53e-05	2.8
p_h (pipeline0_0)	3.37e-06	3.77e-06	7.96e-06	1.51e-05	1.2
clk_gate_C301 (SNPS_CLOCK_GATE_HIGH_top)	6.64e-07	0.000	4.91e-08	7.13e-07	0.1
p2_1 (pipeline2_0)	3.66e-05	4.67e-06	1.93e-06	4.32e-05	3.4
p0m (pipelinem_0)	9.66e-07	2.78e-08	5.16e-08	1.05e-06	0.1
n32 (normal32)	1.58e-06	6.57e-06	5.76e-06	1.39e-05	1.1
add_32 (normal32_DW01_add_2)	0.000	0.000	1.42e-06	1.42e-06	0.1
a1 (adder_0)	2.65e-05	6.40e-05	1.63e-05	1.07e-04	8.4
r328 (adder_0_DW01_add_0)	5.93e-06	5.53e-06	2.38e-06	1.38e-05	1.1
r329 (adder_0_DW01_sub_0)	6.53e-06	6.60e-06	2.89e-06	1.60e-05	1.3

- with clock gate

Hierarchy	Int Power	Switch Power	Leak Power	Total Power	%
top	5.21e-04	5.46e-05	2.17e-04	7.92e-04	100.0
a1_h (adder_2)	0.000	0.000	1.41e-05	1.41e-05	1.8
r328 (adder_2_DW01_add_0)	0.000	0.000	2.53e-06	2.53e-06	0.3
r329 (adder_2_DW01_sub_0)	0.000	0.000	2.97e-06	2.97e-06	0.4
r330 (adder_2_DW_cmp_0)	0.000	0.000	5.35e-07	5.35e-07	0.1
a2_h (adder_1)	0.000	0.000	1.51e-05	1.51e-05	1.9
r328 (adder_1_DW01_add_0)	0.000	0.000	2.12e-06	2.12e-06	0.3
r329 (adder_1_DW01_sub_0)	0.000	0.000	2.78e-06	2.78e-06	0.4
r330 (adder_1_DW_cmp_0)	0.000	0.000	5.35e-07	5.35e-07	0.1
a3_h (adder1_1)	0.000	0.000	1.52e-05	1.52e-05	1.9
r328 (adder1_1_DW01_add_0)	0.000	0.000	2.14e-06	2.14e-06	0.3
r329 (adder1_1_DW01_sub_0)	0.000	0.000	3.36e-06	3.36e-06	0.4
r330 (adder1_1_DW_cmp_0)	0.000	0.000	7.43e-07	7.43e-07	0.1
p0 (pipeline0_1)	1.22e-04	5.02e-06	6.34e-06	1.33e-04	16.8
p2_1_h (pipeline2_1)	3.01e-05	1.23e-06	1.76e-06	3.31e-05	4.2
p1 (pipeline1_0)	2.64e-05	1.04e-06	1.49e-06	2.89e-05	3.7
p2 (pipeline1_7)	2.62e-05	1.04e-06	1.48e-06	2.87e-05	3.6
p3 (pipeline1_6)	2.59e-05	1.04e-06	1.42e-06	2.84e-05	3.6
p4 (pipeline1_5)	2.66e-05	1.04e-06	1.53e-06	2.92e-05	3.7
p_h (pipeline0_0)	1.22e-04	5.02e-06	6.35e-06	1.33e-04	16.8
clk_gate_C301 (SNPS_CLOCK_GATE_HIGH_top)	9.44e-07	0.000	5.23e-08	9.96e-07	0.1
p2_1 (pipeline2_0)	3.03e-05	1.23e-06	1.76e-06	3.33e-05	4.2
p0m (pipelinem_0)	8.49e-07	3.70e-08	5.18e-08	9.38e-07	0.1
n32 (normal32)	3.01e-06	1.80e-05	5.81e-06	2.68e-05	3.4
add_32 (normal32_DW01_add_2)	4.49e-07	1.48e-06	1.43e-06	3.36e-06	0.4
a1 (adder_0)	0.000	0.000	1.53e-05	1.53e-05	1.9



3. Manual (1/2)



gate_level_TB.v

```
// prime time
initial begin
    $dumpvars();
    $dumpfile("DP_4_wave.vcd");

    $sdf_annotate("/home/m:          LU/HW3_ALU/FLP_DP_4/gate/delay/DP_4_syn.sdf", test_module);
end
```

- verilog (gate level netlist)
- ddc (save your design)
- sdc (Synopsys Design Constraints)
- sdf (Standard Delay Format)
- vcd (waveform file)

3. Manual (2/2)

```
#####################################################################
# power analysis
#####################################################################
# source saifmap.ptpx.tcl
# report_name_mapping
set Top DP_4
set power_enable_analysis true
set power_analysis_mode averaged
set power_report_leakage_breakdowns true
# gate-level .v
read_verilog /home/          /back_up/          'ALU/HW3_ALU/FLP_DP_4/gate/delay/DP_4_syn.v
#rtl top module name
current_design $Top
link

read_sdc /home/          /back_up/          'ALU/HW3_ALU/FLP_DP_4/gate/delay/DP_4_syn.sdc
read_sdf /home/          'back_up/          'ALU/HW3_ALU/FLP_DP_4/gate/delay/DP_4_syn.sdf
check_timing
update_timing

read_vcd -strip_path testbench/test_module /home/          /back_up/m          LU/HW3_ALU/FLP_DP_4/postsim/DP_4_wave.vcd

check_power
update_power

report_power -hierarchy > report_${Top}_power_average_vcd.rpt

#quit
exit
```

top module name of your design

file path of gate-level netlist

gate_level_TB.v

```
module testbench();
    integer file_32, file_16, file_ans32, file_ans16;

    reg CLK = 0;
    reg RST = 0;
    reg precision = 0;
    reg     [31:0] data32[0:`DATA_NUM-1];
    reg     [31:0] data16[0:`DATA_NUM-1];
    reg     [63:0] temp_answer;
    reg     [10:0] exp;
    reg     [31:0] input_a1, input_a2, input_a3, input_a4, input_b1, input_b2, inp
    wire     [31:0] answer;
    reg     [31:0]  answer_shift[0:`PIPE];  // answer shift reg
    reg            MODE;
    wire    [31:0] outcome;
    reg stop;
    real real_a1, real_a2, real_a3, real_a4, real_b1, real_b2, real_b3, real_b4, r

    DP_4 test_module( .clk(CLK), .rst(RST), .precision(precision), .a1(input_a1),
                      .c(outcome));
```

END